ECE250: Signals & Systems
Mid-Semester Exam

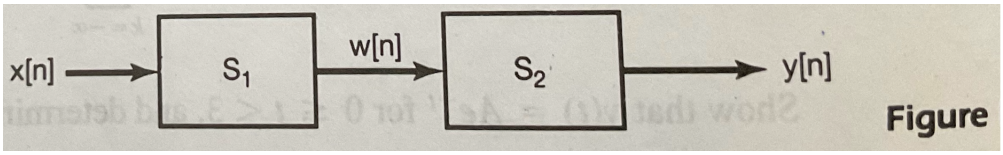
Date: 26th September, 2025 **Total Marks:** 24 **Duration:** 1 hour

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Note:

1. Please provide proper mathematical justifications with your answers. No marks will be awarded without a valid justification.
2. Institute Plagiarism policy are strictly applicable.

[Q1] [CO1, CO2]: Consider the cascade of the two systems S_1 and S_2, as depicted in the figure below.



Figure

- S_1: $w[n] = \frac{1}{2}w[n-1] + x[n]$
- S_2: $y[n] = \alpha y[n-1] + \beta w[n]$

The difference equation relating $w[n]$ and $y[n]$ is:

$$S_1 : y[n] = -\frac{1}{8}y[n-2] + \frac{3}{4}y[n-1] + x[n]$$

(a) Determine α and β. **[4 Marks]**

(b) Find the impulse response of the cascade connection of S_1 and S_2. **[4 Marks]**

[Q2] [CO1, CO2]: Given the impulse response of an LTI system as below. Determine whether this system is causal and/or stable. Justify your answer. **[4 Marks]**

$$h(t) = e^{-6|t|}$$

[Q3] [CO1]: Determine whether or not each of the following time-domain signals is periodic. If the signal is periodic, determine its fundamental time period.

(i) $x(t) = 3\cos(4t + \frac{\pi}{3})$ **[3 Marks]**

(ii) $x[n] = \cos(\frac{\pi}{2}n)\cos(\frac{\pi}{4}n)$ **[3 Marks]**

[Q4] [CO1, CO2, CO3]: Given the description of a continuous-time periodic signal $x(t)$ in one time-period as below.

$$x(t) = \begin{cases} 1.5, & 0 \le t < 1 \\ -1.5, & 1 \le t < 2 \end{cases}$$

a) Find the fundamental frequency. **[1 Marks]**
b) Find the Fourier series coefficients a_k for the above signal. **[5 Marks]**

Bonus Question

[CO1, CO2, CO3] Please complete the 2-part question below.

(i) Write the definition and properties of a Vector Space. **[2 Marks]**

(ii) Prove that a collection of all continuous time signals periodic with time period T form a vector space over the field of complex numbers. **[2 Marks]**